

Build and debug 'Linux Hello World' with Vitis Unified

- Want to run/debug the application on our ZedBoard
- Will use the 'linux hello world' example project that comes with Vitis

Using Vitis Unified 2024.2:

1. Set workspace to 'Projects/VitisZedLinuxWs' (brand new folder)
2. In 'Embedded Development' panel, click 'Create Platform Component'

The screenshot shows the 'Create Platform Component' dialog with the 'Name and Location' tab selected. The breadcrumb navigation is 'Name and Location > Flow > OS and Processor > Summary'. The 'Name and Location' section instructs the user to choose a name and specify a directory. The 'Component name' field contains 'platform_zed_board_linux'. The 'Component location' dropdown shows 'C:\Projects\VitisZedLinuxWs' with a 'Browse' button next to it. Below this, it states 'Component will be created at C:\Projects\VitisZedLinuxWs\platform_zed_board_linux'. A note at the bottom says: 'Note: Avoid using excessively deep paths, as you may encounter issues on Windows systems due to the character limit.' At the bottom of the dialog are 'Cancel' and 'Next' buttons.

Create Platform Component

Name and Location > Flow > OS and Processor > Summary

Name and Location

Choose a name for your component and specify a directory where the component data files will be stored.

Component name: platform_zed_board_linux

Component location: C:\Projects\VitisZedLinuxWs Browse

Component will be created at C:\Projects\VitisZedLinuxWs\platform_zed_board_linux

Note: Avoid using excessively deep paths, as you may encounter issues on Windows systems due to the character limit.

Cancel Next

The screenshot shows the 'Create Platform Component' dialog with the 'Flow' tab selected. The breadcrumb navigation is 'Name and Location > Flow > OS and Processor > Summary'. The 'Select Platform Creation Flow' section instructs the user to create a platform component by selecting the hardware design and add software domains. There are two radio buttons: 'Hardware Design' (selected) and 'Existing Platform'. Below this, the 'Hardware Design (XSA) For Implementation' section has a '> Advanced Options' link. A dropdown menu 'Select hardware design' is open, showing a list of hardware designs: vck190, vmk180, zcu102, zcu106, zc702, zc706, and zed. Each design has a 'Fixed' status indicator to its right. A 'Browse' button is to the right of the dropdown. At the bottom of the dialog are 'Cancel', 'Back', and 'Next' buttons.

Create Platform Component

Name and Location > Flow > OS and Processor > Summary

Select Platform Creation Flow

Create a platform component by selecting the hardware design and add software domains.

☒ Hardware Design ☐ Existing Platform

Hardware Design (XSA) For Implementation

> Advanced Options

Select hardware design

Select hardware design	
vck190	Fixed
vmk180	Fixed
zcu102	Fixed
zcu106	Fixed
zc702	Fixed
zc706	Fixed
zed	Fixed

Browse

Cancel Back Next

Create Platform Component

Name and Location > Flow > OS and Processor > Summary

Select Platform Creation Flow

Create a platform component by selecting the hardware design and add software domains.

☒ Hardware Design

☐ Existing Platform

Hardware Design (XSA)
For Implementation

zed

Browse

Advanced Options

SDT Source Repo

Browse

Board DTSi

User DTSi

Browse

DT ZOCL

☐

Cancel

Back

Next

Create Platform Component

Name and Location > Flow > OS and Processor > Summary

Select Operating System and Processor

Specify the details for the initial domain to be added to the platform component. The platform can be modified later to add new domains or change settings by opening the platform editor.

Operating system:

linux

Processor:

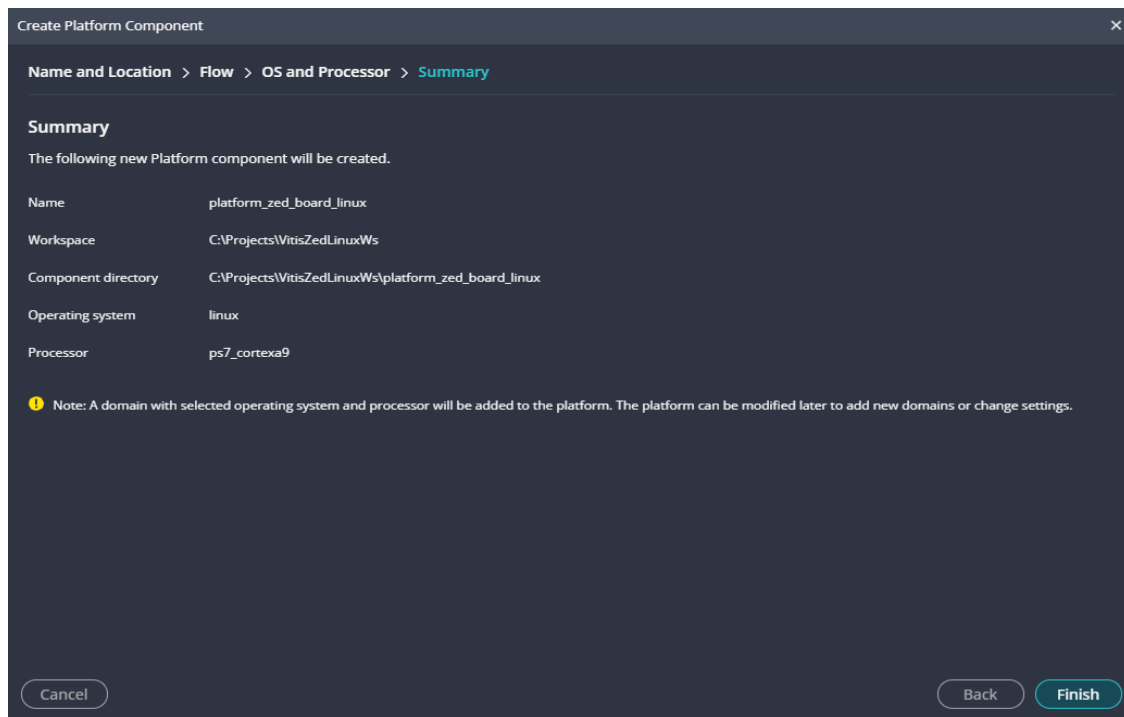
ps7_cortexa9

☒ Generate Boot artifacts

Cancel

Back

Next



Output console:

```
18:01:33 INFO : updated workspace path: C:\Projects\VitisZedLinuxWs
18:01:35 INFO : Successfully created repository data at
C:\Users\wlaba\AppData\Local\Temp\rigel_lopper_wlaba15563819171094951191
18:01:35 INFO : cmd.exe, /C, C:\Xilinx\Vitis\2024.2\tps\win64\lopper-
1.1.0\env\Scripts\activate.bat && python
C:\Xilinx\Vitis\2024.2\data\embeddedsw\scripts\pyesw\get_template_data.py -d
C:\Users\wlaba\AppData\Local\Temp\rigel_lopper_wlaba11976232445309139885 -r
C:\Users\wlaba\AppData\Local\Temp\rigel_lopper_wlaba15563819171094951191\.repo.yaml
&& C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Scripts\deactivate.bat
18:01:36 INFO : Embedded Template List generated successfully
18:01:36 INFO : cmd.exe, /C, C:\Xilinx\Vitis\2024.2\tps\win64\lopper-
1.1.0\env\Scripts\activate.bat && python
C:\Xilinx\Vitis\2024.2\data\embeddedsw\scripts\pyesw\repo.py -st
C:\Xilinx\Vitis\2024.2\data\embeddedsw && C:\Xilinx\Vitis\2024.2\tps\win64\lopper-
1.1.0\env\Scripts\deactivate.bat
18:01:38 INFO : Successfully created repository data at
C:\Projects\VitisZedLinuxWs\_ide\.wsdata
18:01:38 INFO : cmd.exe, /C, C:\Xilinx\Vitis\2024.2\tps\win64\lopper-
1.1.0\env\Scripts\activate.bat && python
C:\Xilinx\Vitis\2024.2\data\embeddedsw\scripts\pyesw\get_template_data.py -d
C:\Users\wlaba\AppData\Local\Temp\rigel_lopper_wlaba8550755919004480838 -r
C:\Projects\VitisZedLinuxWs\_ide\.wsdata\.repo.yaml &&
C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Scripts\deactivate.bat
18:01:39 INFO : Embedded Template List generated successfully
18:40:59 INFO : Found no platform with name 'zed' in install repositories

18:40:59 INFO : Install SDT for given XSA copied Successfully
```

```
18:40:59 INFO : cmd.exe, /C, C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Scripts\activate.bat && lopper --enhanced --werror -f -O C:\Projects\VitisZedLinuxWs\rigel_lopper -i C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Lib\site-packages\lopper\lops\lop-cpu-oslist.dts C:\Projects\VitisZedLinuxWs\rigel_lopper_temp_platform\zed\hw\sdt\system-top.dts && C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Scripts\deactivate.bat
```

```
18:41:01 INFO : CPU List generated successfully
```

```
18:47:13 INFO : Platform platform_zed_board_linux creation started.
```

```
18:47:13 INFO : Lopper command for cpu List generation: lopper --enhanced --werror -f -O C:\Projects\VitisZedLinuxWs\rigel_lopper -i C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Lib\site-packages\lopper\lops\lop-cpu-oslist.dts C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\hw\sdt\system-top.dts
```

```
18:47:13 INFO : cmd.exe, /C, C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Scripts\activate.bat && lopper --enhanced --werror -f -O C:\Projects\VitisZedLinuxWs\rigel_lopper -i C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Lib\site-packages\lopper\lops\lop-cpu-oslist.dts C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\hw\sdt\system-top.dts && C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Scripts\deactivate.bat
```

```
18:47:14 INFO : CPU List generated successfully
```

```
18:47:14 INFO : created .gitignore file for the project C:\Projects\VitisZedLinuxWs\platform_zed_board_linux
```

```
18:47:14 INFO : ZYNQ: Using the QEMU args from install at : C:\Xilinx\Vitis\2024.2\data\emulation\platforms\zynq\sw\z9_linux\qemu\qemu_args.txt
```

```
18:47:14 INFO : ZYNQ: Using the QEMU args from install at : C:\Xilinx\Vitis\2024.2\data\emulation\platforms\zynq\sw\z9_standalone\qemu\qemu_args.txt
```

```
18:47:14 INFO : ZYNQ: Using the QEMU args from install at : C:\Xilinx\Vitis\2024.2\data\emulation\platforms\zynq\sw\z9_linux\qemu\qemu_args.txt
```

```
18:47:14 INFO : Domain linux_ps7_cortexa9 added successfully.
```

```
18:47:14 INFO : lopper command to generate BSP :python C:\Xilinx\Vitis\2024.2\data\embeddedsw/scripts/pyesw/create_bsp.py -w C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\zynq_fsbl\zynq_fsbl_bsp -p ps7_cortexa9_0 -o standalone -s C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\hw\sdt\system-top.dts -t zynq_fsbl -r C:\Projects\VitisZedLinuxWs\ide\wsdata\repo.yaml
```

```
18:47:14 INFO : cmd.exe, /C, C:\Xilinx\Vitis\2024.2\tps\win64\lopper-
```

```
1.1.0\env\Scripts\activate.bat && python
C:\Xilinx\Vitis\2024.2\data\embeddedsw\scripts\pyesw\create_bsp.py -w
C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\zynq_fsbl\zynq_fsbl_bsp -p
ps7_cortexa9_0 -o standalone -s
C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\hw\sdt\system-top.dts -t
zynq_fsbl -r C:\Projects\VitisZedLinuxWs\_ide\.wsdata\.repo.yaml &&
C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Scripts\deactivate.bat
```

```
18:47:29 INFO : Successfully created Domain at
C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\zynq_fsbl\zynq_fsbl_bsp
```

```
18:47:29 INFO : lopper command to create baremetal application :python
C:\Xilinx\Vitis\2024.2\data\embeddedsw\scripts\pyesw\create_app.py -s
C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\zynq_fsbl -t zynq_fsbl -d
C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\zynq_fsbl\zynq_fsbl_bsp -n
fsbl -r C:\Projects\VitisZedLinuxWs\_ide\.wsdata\.repo.yaml
```

```
18:47:29 INFO : cmd.exe, /C, C:\Xilinx\Vitis\2024.2\tps\win64\lopper-
1.1.0\env\Scripts\activate.bat && python
C:\Xilinx\Vitis\2024.2\data\embeddedsw\scripts\pyesw\validate_bsp.py -t zynq_fsbl
-d C:\Projects\VitisZedLinuxWs\platform_zed_board_linux\zynq_fsbl\zynq_fsbl_bsp -r
C:\Projects\VitisZedLinuxWs\_ide\.wsdata\.repo.yaml &&
C:\Xilinx\Vitis\2024.2\tps\win64\lopper-1.1.0\env\Scripts\deactivate.bat
```

```
18:47:32 INFO : Successfully validated template zynq_fsbl
```

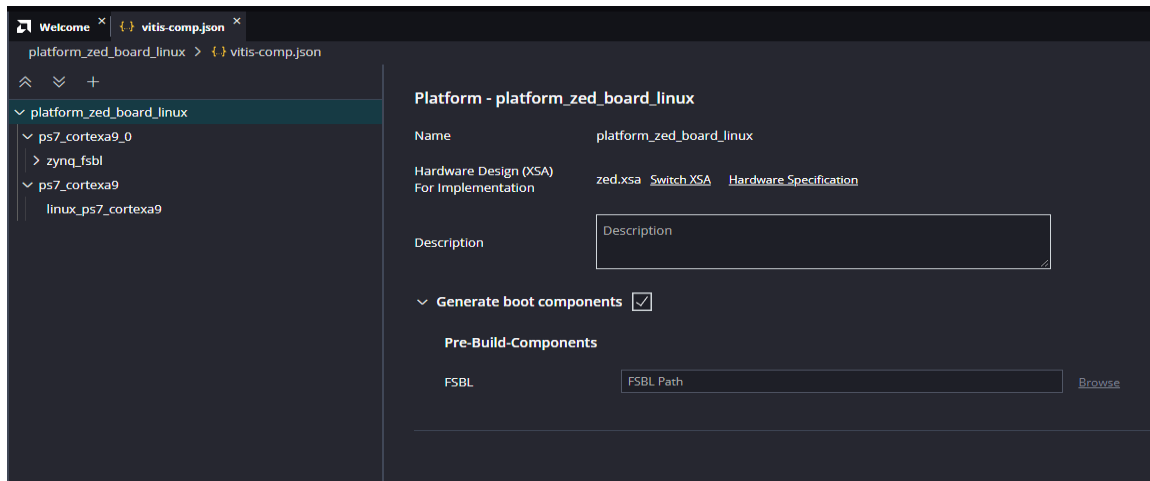
```
18:47:37 INFO : Successfully Created Application sources at
C:/Projects/VitisZedLinuxWs/platform_zed_board_linux/zynq_fsbl
```

```
18:47:37 INFO : Platform FSBL Boot domain added successfully.
```

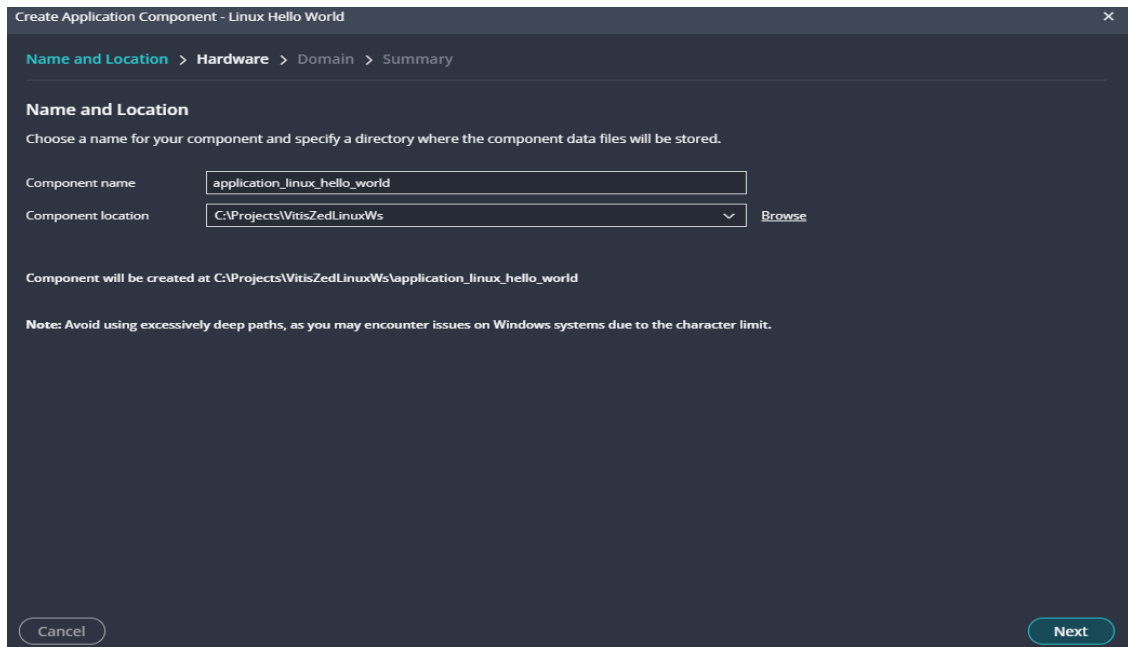
```
18:47:38 INFO : Platform creation finished successfully.
```

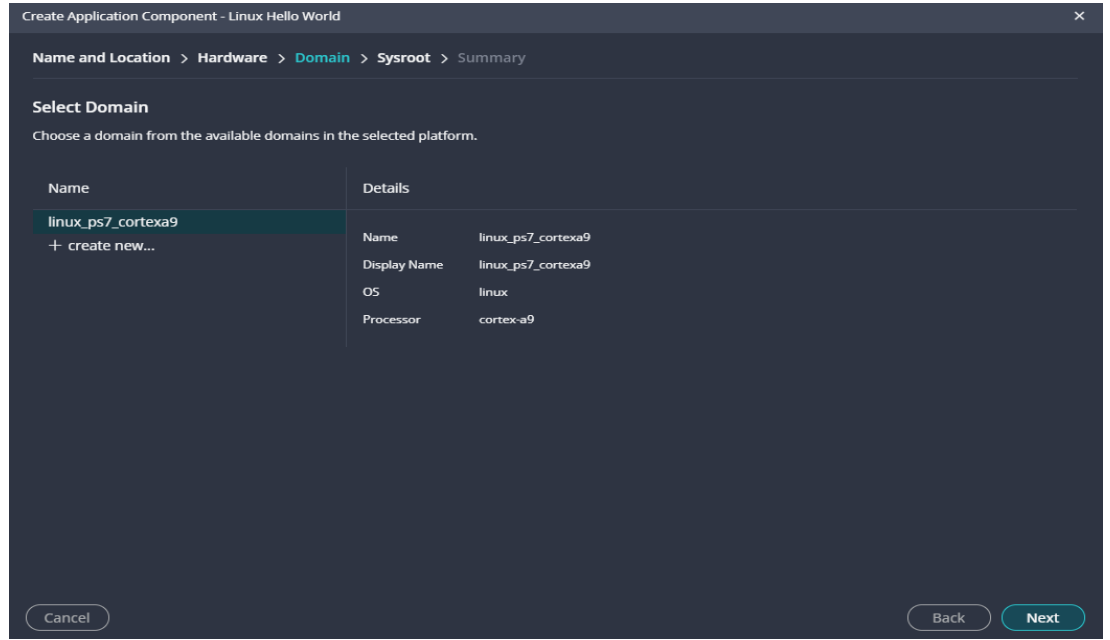
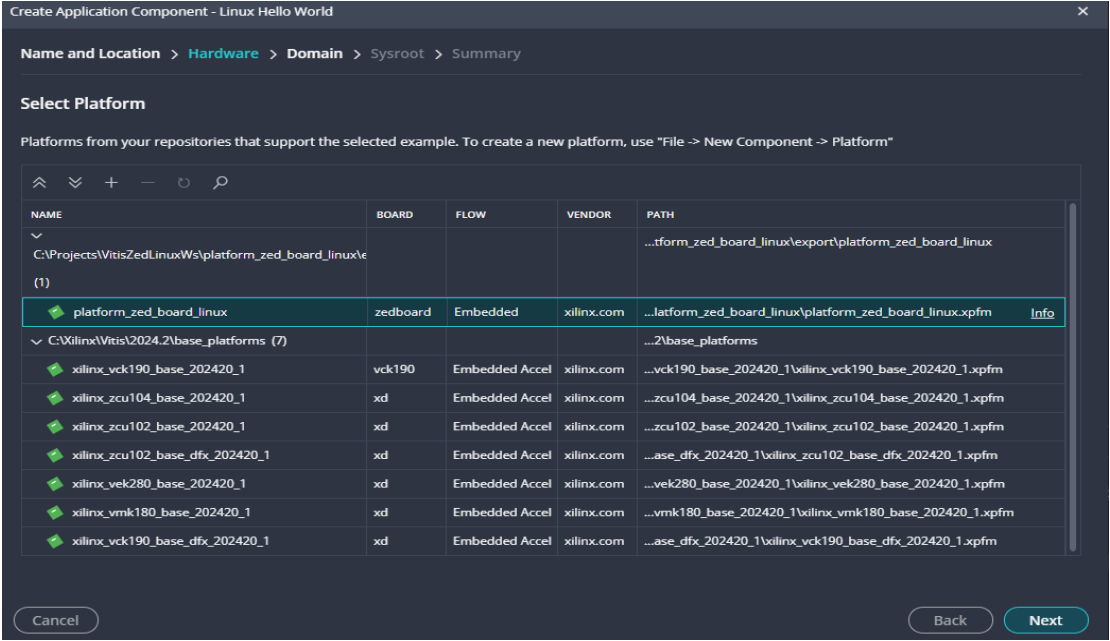
```
18:47:38 INFO : Generating Export directory
```

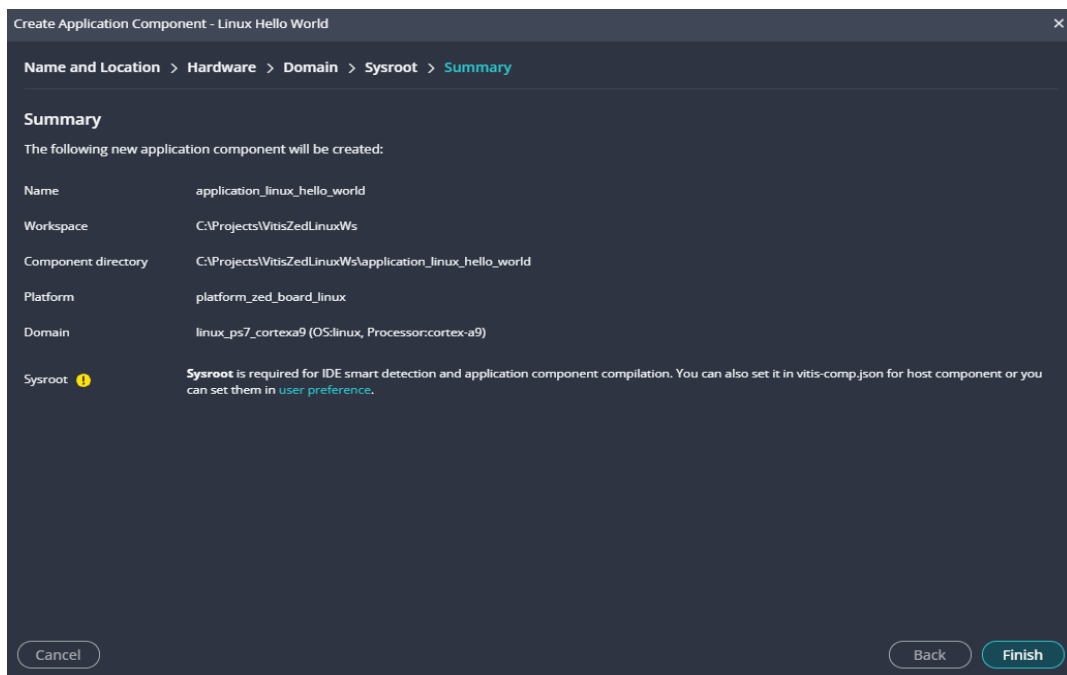
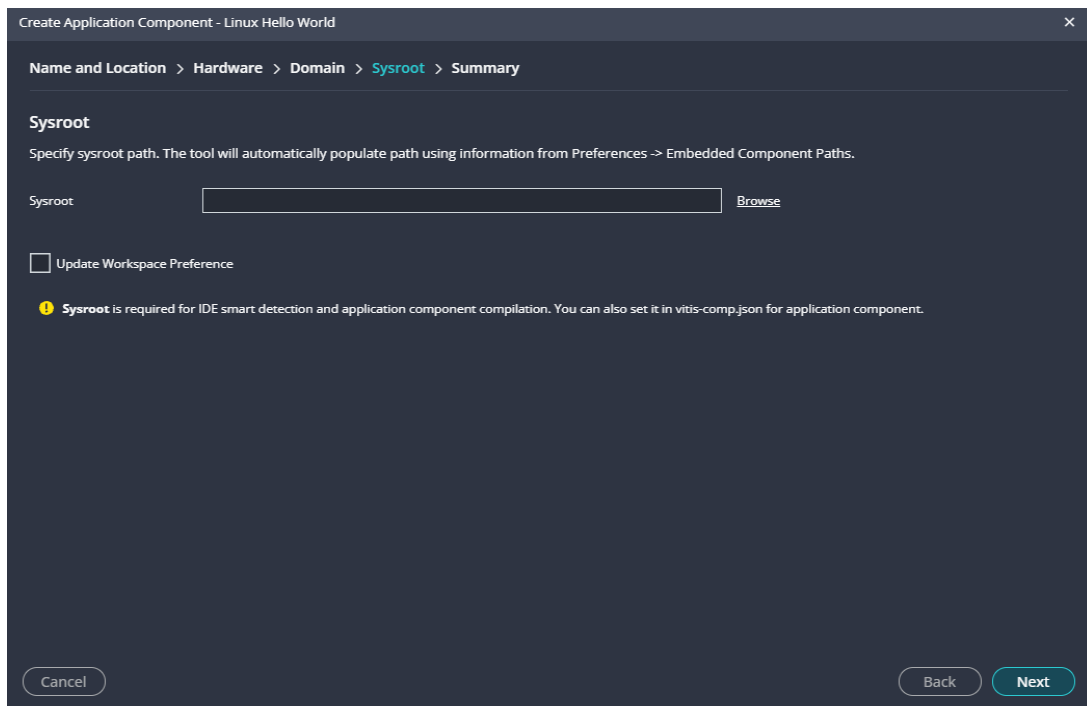
```
18:47:38 INFO : Generated platform metadata for creating application(s) based on
platform platform_zed_board_linux.
```



- Click on Examples Icon and select '+' sign to right of 'Linux Hello World' to create 'New Application Component'





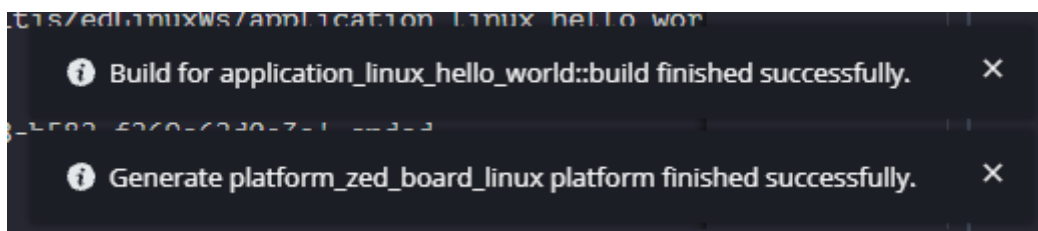
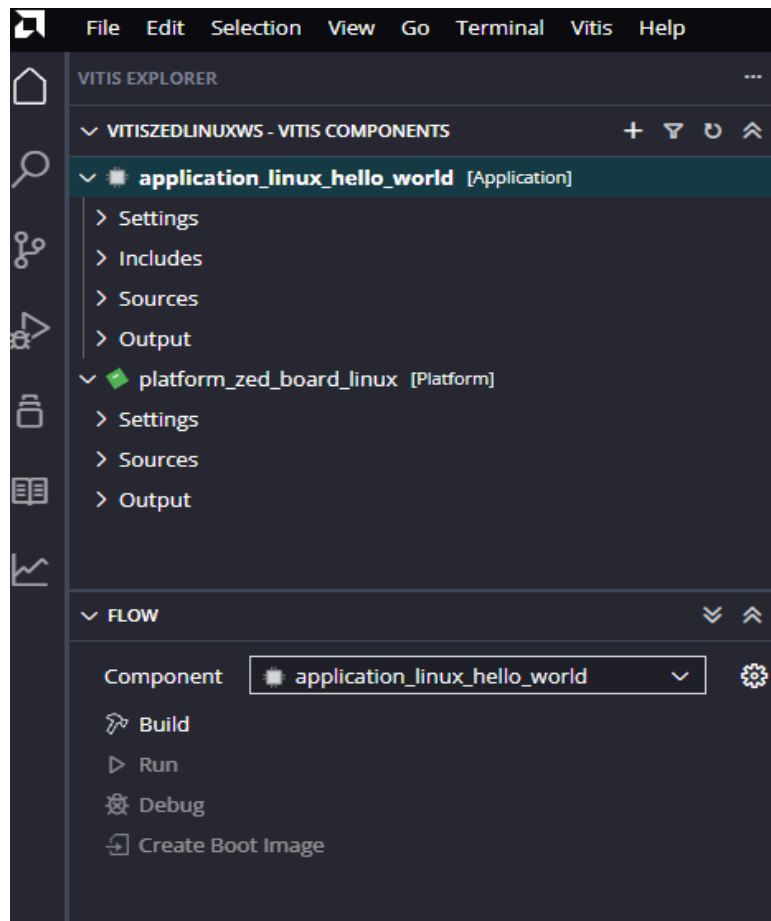


Output Console:

```
19:01:42 INFO : Found compiledb at C:\Xilinx\Vitis\2024.2\vitis-server\scripts\compiledb
```

```
19:01:45 INFO : Generated compile_commands.json at C:\Projects\VitisZedLinuxWs\application_linux_hello_world\_ide
```

4. Building hello world project:



Console Output:

```
-----
[7/16/2025, 7:07:20 PM]: Build for application_linux_hello_world::build with id
'5167207d-f0ec-4dc8-b582-f269e62d0a3a' started.
-----
-- The C compiler identification is GNU 13.3.0
-- The CXX compiler identification is GNU 13.3.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: C:/Xilinx/Vitis/2024.2/gnu/aarch32/nt/gcc-arm-
linux-gnueabi/bin/arm-linux-gnueabihf-gcc.exe - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: C:/Xilinx/Vitis/2024.2/gnu/aarch32/nt/gcc-arm-
linux-gnueabi/bin/arm-linux-gnueabihf-g++.exe - skipped
-- Detecting CXX compile features
```

```

-- Detecting CXX compile features - done
-- Configuring done
-- Generating done
-- Build files have been written to:
C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build
C:/Xilinx/Vitis/2024.2/tps/win64/cmake-3.24.2/bin/cmake.exe
-SC:/Projects/VitisZedLinuxWs/application_linux_hello_world
-BC:/Projects/VitisZedLinuxWs/application_linux_hello_world/build --check-build-
system CMakeFiles/Makefile.cmake 0
C:/Xilinx/Vitis/2024.2/tps/win64/cmake-3.24.2/bin/cmake.exe -E
cmake_progress_start
C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build/CMakeFiles
C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build//CMakeFiles/progres
s.marks
C:/ti/xdctools_3_23_04_60/gmake.exe -f CMakeFiles/Makefile2 all
gmake.exe[1]: Entering directory
`C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build'
C:/ti/xdctools_3_23_04_60/gmake.exe -f
CMakeFiles/application_linux_hello_world.elf.dir/build.make
CMakeFiles/application_linux_hello_world.elf.dir/depend
gmake.exe[2]: Entering directory
`C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build'
C:/Xilinx/Vitis/2024.2/tps/win64/cmake-3.24.2/bin/cmake.exe -E cmake_depends "Unix
Makefiles" C:/Projects/VitisZedLinuxWs/application_linux_hello_world
C:/Projects/VitisZedLinuxWs/application_linux_hello_world
C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build
C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build
C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build/CMakeFiles/applicat
ion_linux_hello_world.elf.dir/DependInfo.cmake --color=
gmake.exe[2]: Leaving directory
`C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build'
C:/ti/xdctools_3_23_04_60/gmake.exe -f
CMakeFiles/application_linux_hello_world.elf.dir/build.make
CMakeFiles/application_linux_hello_world.elf.dir/build
gmake.exe[2]: Entering directory
`C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build'
[ 50%] [32mBuilding C object
CMakeFiles/application_linux_hello_world.elf.dir/src/helloworld.c.obj[0m
C:/Xilinx/Vitis/2024.2/gnu/aarch32/nt/gcc-arm-linux-gnueabi/bin/arm-linux-
gnueabihf-gcc.exe -DVITIS_MAJOR_VERSION="" -DVITIS_MINOR_VERSION=""
-DVITIS_VERSION="" -g -Wall -Wextra -O0 -g3 -MD -MT
CMakeFiles/application_linux_hello_world.elf.dir/src/helloworld.c.obj -MF
CMakeFiles/application_linux_hello_world.elf.dir/src/helloworld.c.obj.d -o
CMakeFiles/application_linux_hello_world.elf.dir/src/helloworld.c.obj -c
C:/Projects/VitisZedLinuxWs/application_linux_hello_world/src/helloworld.c
[100%] [32m[1mlinking C executable application_linux_hello_world.elf[0m
C:/Xilinx/Vitis/2024.2/gnu/aarch32/nt/gcc-arm-linux-gnueabi/bin/arm-linux-
gnueabihf-gcc.exe
"CMakeFiles/application_linux_hello_world.elf.dir/src/helloworld.c.obj" -o
application_linux_hello_world.elf
gmake.exe[2]: Leaving directory

```

```
`C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build'
[100%] Built target application_linux_hello_world.elf
gmake.exe[1]: Leaving directory
`C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build'
C:/Xilinx/Vitis/2024.2/tps/win64/cmake-3.24.2/bin/cmake.exe -E
cmake_progress_start
C:/Projects/VitisZedLinuxWs/application_linux_hello_world/build/CMakeFiles 0
Build Finished successfully
-----
[7/16/2025, 7:07:24 PM]: Build for application_linux_hello_world::build with id
'5167207d-f0ec-4dc8-b582-f269e62d0a3a' ended.
```

- Looks like we successfully created/built a ZedBoard platform and Linux Hello World application using Vitis!
- Next, we are going to try to launch a debug session from Vitis